

CHAPTER 8

Creep Behavior of Particle-Strengthened Alloys

Contents

1. Introduction	167
2. Small Volume-Fraction Particles Coherent and Incoherent with the Matrix with Small Aspect Ratios	168
2.1 Introduction and Theory	168
2.2 Local and General Climb of Dislocations over Obstacles	172
2.3 Detachment Model	175
2.4 Constitutive Relationships	180
2.5 Microstructural Effects	184
2.5.1 Transient Creep Behavior and Dislocation Structure	184
2.5.2 Effect of Volume Fraction	186
2.5.3 Grain Size Effects	187
2.6 Coherent Particles	187

1. INTRODUCTION

This chapter will discuss the behavior of two types of materials that have creep properties enhanced by second phases. These materials contain particles with square or spherical aspect ratios that are both coherent and incoherent with the matrix, and are of relatively low volume fractions. This chapter is a review of work in this area, but it must be recognized that other reviews have been published and this chapter reflects the particularly high-quality reviews by Reppich and coworkers [538–540], Arzt [541,542], and others [543–549]. It should also be mentioned that this chapter will emphasize those cases where there are relatively wide separations between the particles, or stated another way, the volume fractions of the precipitate discussed here are nearly always less than 30% and usually less than 10%. This contrasts with the case of some γ/γ' alloys where the precipitate, γ' , occupies a substantial volume fraction of the alloys; these are discussed in Chapter 11.

2. SMALL VOLUME-FRACTION PARTICLES COHERENT AND INCOHERENT WITH THE MATRIX WITH SMALL ASPECT RATIOS

2.1 Introduction and Theory

It is well known that second-phase particles provide enhanced strength at lower temperatures, and there have been numerous discussions on the source of this strength. A review of low-temperature strengthening by second-phase particles was published by Reppich [539]. Although a discussion of the mechanisms of lower-temperature second-phase strengthening is outside the scope of this chapter, it should be mentioned that the strength by particles has been believed to be provided in two somewhat broad categories of strengthening, Friedel cutting or Orowan bypassing. Basically, the former involves coherent particles and the flow stress of the alloy is governed by the stress required for the passage of the dislocation through a particle. The Orowan stress is determined by the bypass stress based on an Orowan loop mechanism. In the case of oxide dispersion strengthened (ODS) alloys, in which the particles are incoherent, the low temperature yield stress is reasonably predicted by the Orowan loop mechanism [539,550]. The Orowan bowing stress is approximated by the classic equation:

$$\tau_{\text{or}} = Gb/L = \left(\frac{2T_d}{bL} \right) \quad (150)$$

where τ_{or} is the bowing stress, L is the average separation between particles, and T_d is the dislocation line tension. This equation, of course, assumes that the elastic strain energy of a dislocation can be estimated by $Gb^2/2$, which, though reasonable, is not firmly established [88].

This chapter discusses how the addition of second phases leads to enhanced strength (creep resistance) at elevated temperatures. This discussion is important for at least two reasons. First, as [Figure 80](#) illustrates, the situation at elevated temperature is different than at lower temperatures. This figure illustrates that the yield stress of the single-phase matrix is temperature-dependent, of course, but there is a superimposed strengthening (suggested in the figure to be approximately athermal) by Orowan bowing [550]. It is generally assumed that the bowing process cannot be thermally activated, but the non-shearable particles can be negotiated by climb. At higher temperatures, it is suggested that the flow stress becomes only a fraction of this superimposed stress and an understanding of the

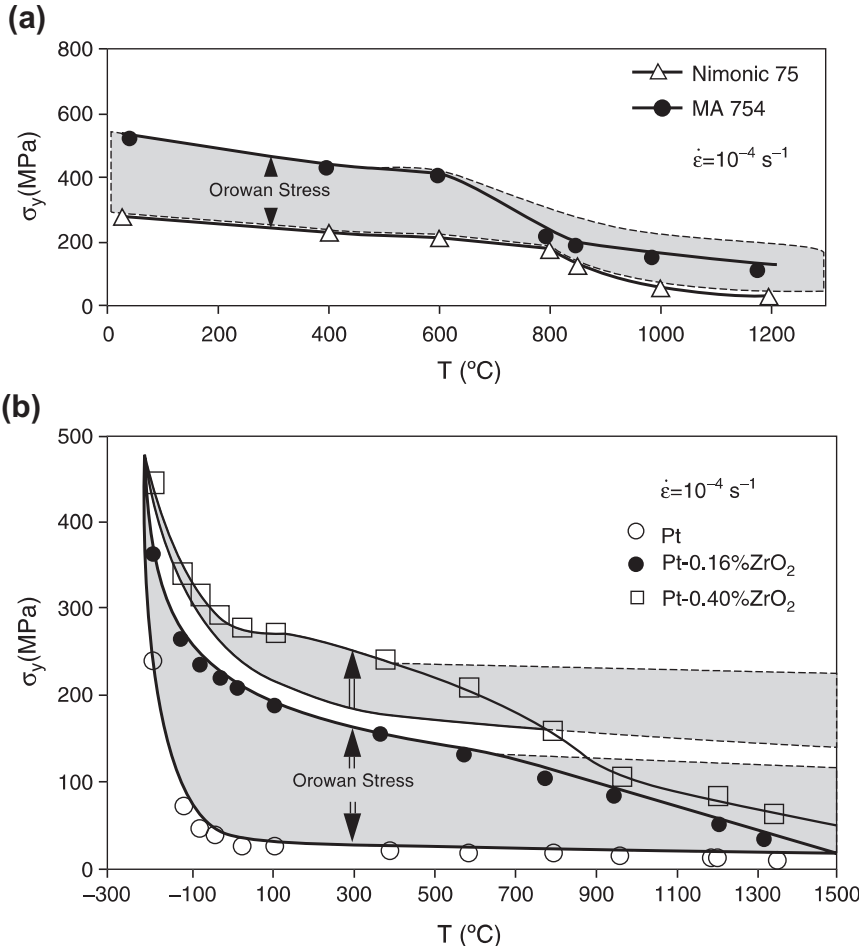


Figure 80 Compressive 0.2% yield stress versus temperature. (Shaded: Orowan stress given as low-temperature yield-stress increment due to oxide dispersoids.) (a) ODS Superalloy MA 754. (b) Pt-based ODS alloys. *From Ref. [539].*

origin is a significant focus of this chapter. Second, in the previous chapters it was illustrated how solute additions, basically obstacles, lead to increased creep strength. There is, essentially, a roughly uniform shifting of the power law, power-law breakdown, and low-stress-exponent regimes to higher stresses. This is evident in Figure 81, where the additions of Mg to Al are described. In the case of alloys with second-phase particles, however, there is often a lack of this uniform shift and sometimes the appearance that many investigators have termed a “threshold stress,” σ_{th} . The intent of this

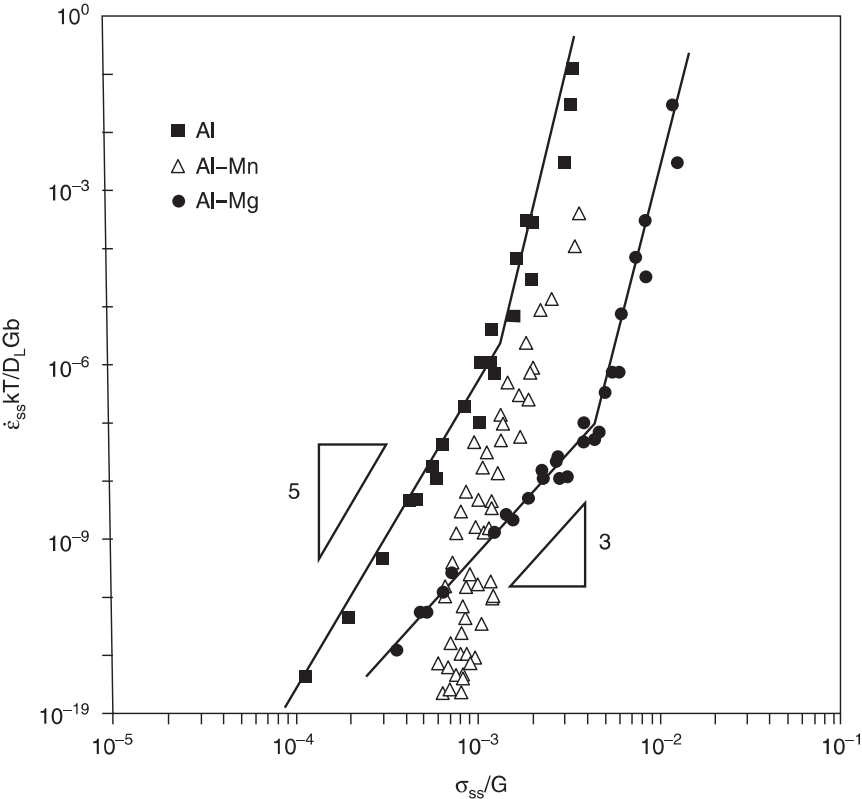


Figure 81 Steady-state relation between strain rate $\dot{\epsilon}$ and flow stress for the alloys of this work compared to literature data from slow tests (Al, Al-Mg, Al-Mn). Adapted from Ref. [551].

term is illustrated in Figure 82, based on the data of Lund and Nix and additional interpretations by Pharr and Nix [552,553]. Figure 82 reflects “classic” particle strengthening by oxide dispersoids (ThO_2) in a Ni-Cr solid solution matrix. These particles, of course, are incoherent with the matrix. The “pure” solid solution alloy behavior is also indicated. Particle strengthening is evident at all steady-state stress levels, but at lower stresses there appears a modulus-compensated stress below which creep does not appear to occur or is at least very slow. Again, this has been termed the threshold stress, σ_{th} . Said another way, there is not a uniform shift of the strengthening on logarithmic axes; there appears a larger fraction of the strength provided by the particles at lower stress (high temperatures) than at higher stresses (lower temperatures). In fact, the concept of a

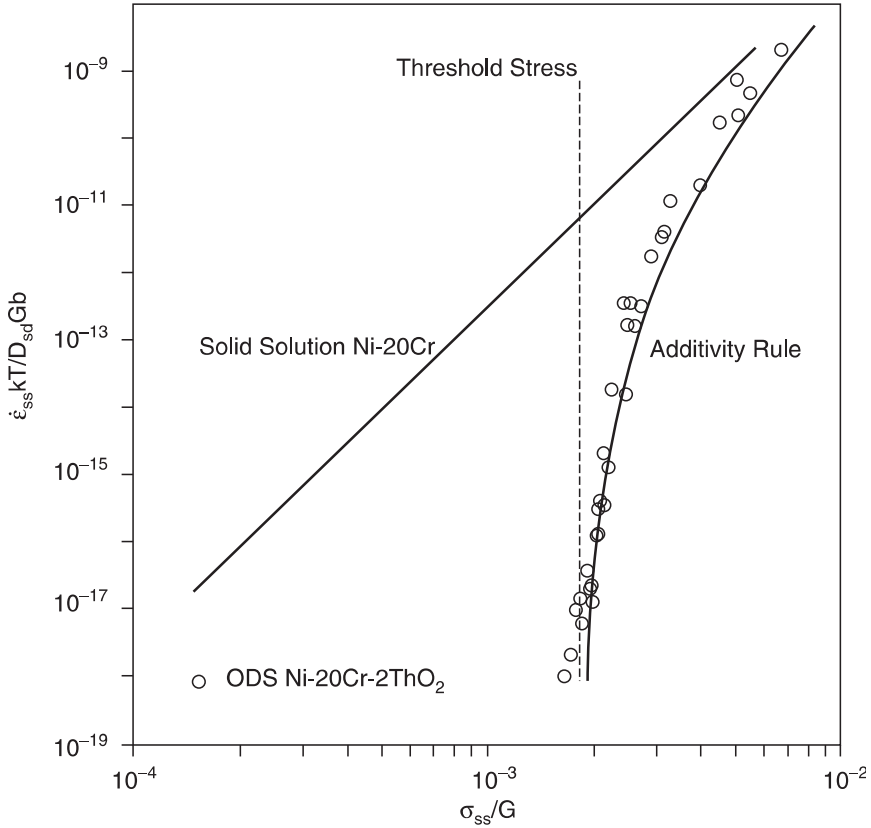


Figure 82 The normalized steady-state creep-rate versus modulus compensated steady-state stress. *Adapted from Refs [552,553].*

threshold stress was probably originally considered one of athermal strengthening. This will be discussed more later, but this “coarse” description identifies an aspect of particle strengthening that appears generally different from dislocation-substructure strengthening and solution strengthening (although the latter, in certain temperature regimes, may have a nearly athermal strengthening character). The threshold stress of Figure 82 is only about half the Orowan bowing stress, suggesting that Orowan bowing may not be the basis of the threshold. Activation energies appear relatively high (greater than lattice self-diffusion of the matrix) as well as the stress exponents being relatively high ($n \gg 5$) in the region where a threshold is apparent.

Particle strengthening is also illustrated in Figure 81, based on a figure from reference [551], where there is, again, a nonuniform shift in the

behavior of the particle-strengthened Al. The figure indicates the classic creep behavior of high-purity Al. Additionally, the behavior of Mg (4.8 wt%)-solute strengthened Al is plotted (there is additionally about 0.05 wt% Fe and Si solute in this alloy). The Mg atoms significantly strengthen the Al. The strengthening may be associated with viscous glide in some temperature ranges in this case. However, an important point is that at higher modulus-compensated stresses, the Al–Mn alloy (strengthened by incoherent Al_6Mn particles) has slightly greater strength than pure Al (there is also an additional 0.05 wt% Fe and Si solute in this alloy). It does not appear to have as high a strength as the solute-strengthened Al–4.8wt% Mg alloy. However, at lower applied stresses (lower strain rates) or higher temperatures, the second-phase strengthened alloy has *higher* strength than both the pure matrix and the solution strengthened alloy. As with the ODS alloy of [Figure 82](#), a threshold behavior is evident. The potential technological advantage of these alloys appears to be provided by this threshold-like behavior. As the temperature is increased, the flow stress does not experience the magnitude of decreases as by the other (e.g., solute and dislocation) strengthening mechanisms.

Basically, the current theories for the threshold stress fall into one of two main categories: a threshold arising due to increased dislocation line length with climb over particles and the detachment stress to remove the dislocation from the particle matrix interface after climb over the particle.

2.2 Local and General Climb of Dislocations over Obstacles

It was presumed long ago, by Ansell and Weertman [554], that dislocation climb allowed for passage at these elevated temperatures and relatively low stresses. The problem with this early climb approach is that the creep rate is expected to have a low stress-dependence with an activation energy equivalent to that of lattice self-diffusion. As indicated in the figures just presented in this chapter, the stress dependence in the vicinity of the threshold is relatively high and the activation energy in this threshold regime can be much higher than that of lattice self-diffusion. More recent analysis has attempted to rationalize the apparent threshold. One of the earlier approaches suggested that for stresses below the cutting, σ_{ct} (relevant for some cases of coherent precipitates) or Orowan bowing stress, σ_{or} , the dislocation must—as Ansell and Weertman originally suggested—climb over the obstacle. This climbing process could imply an increase in dislocation line length and hence total elastic strain energy, which would act as an impediment to plastic flow [555–560]. The schemes by which this has

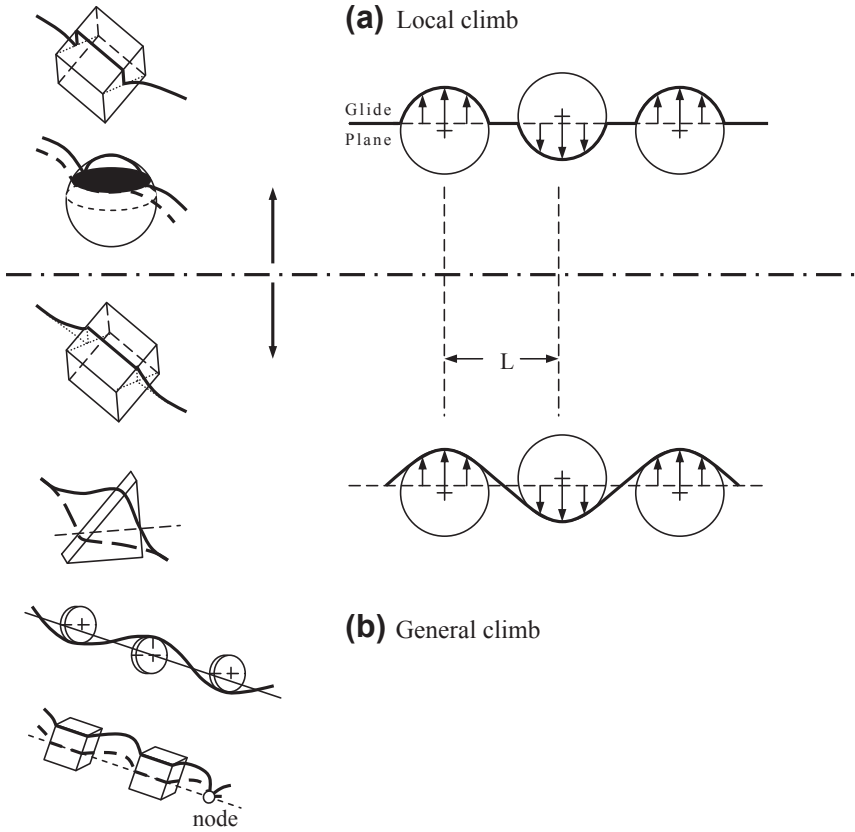


Figure 83 Compilation by Blum and Reppich [538] of models for dislocation climb over second-phase particles.

been suggested are illustrated in Figure 83. Figure 84 shows an edge dislocation climbing, with concomitant slip, over a spherical particle. As the dislocation climbs, work is performed by the applied shear. The total energy change can be described by:

$$dE \cong (Gb^2/2)dL - \tau bLdx - \sigma_n bLdy - dE_{el} \quad (151)$$

The first term is the increase in elastic strain energy associated with the increase in dislocation line length. This is generally the principal term giving rise to the (so-called or apparent) threshold stress. The second term is the work done by the applied stress as the dislocation glides. The third term is the work done by the normal component of the stress as the dislocation climbs. The fourth term accounts for any elastic interaction between the

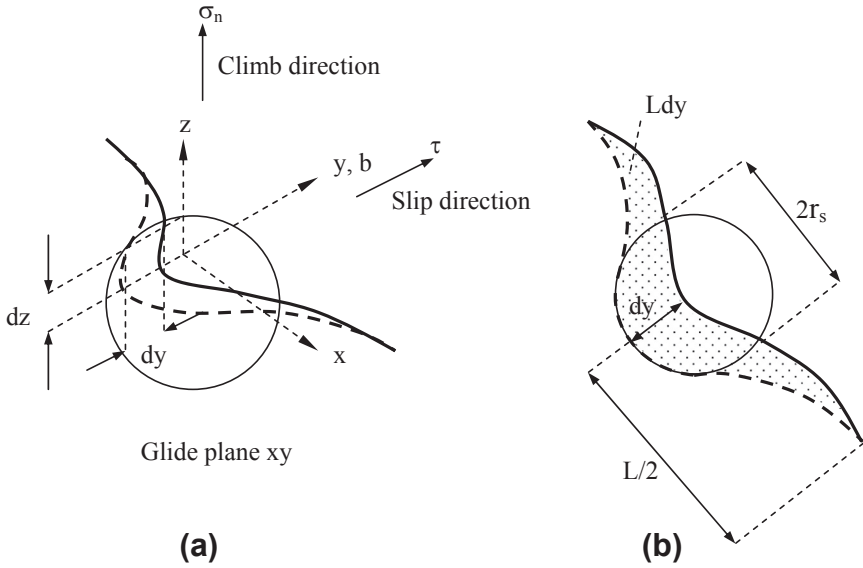


Figure 84 (a) Climb of an edge dislocation over a spherical particle. (b) Top view. *From Ref. [538].*

dislocation and the particle [561], which does not, in its original formulation, appear to include coherency stresses, although this would be appropriate for coherent particles. This equation is often simplified to:

$$dE = (Gb^2/2)dL - \tau bLdx \quad (152)$$

The critical stress for climb of the dislocation over the particle is defined under the condition where:

$$(dE/dx) = 0$$

or:

$$\tau_c = Gb/L(0.5\alpha') \quad (153)$$

Arzt and Ashby defined the α' parameter $= (dL/dx)_{\max}$ as the climb resistance, and τ_c can be regarded as the apparent threshold stress. Estimates have been made of α' by relating the volume fraction of the particles, and the particle diameter to the value of L in Eqn (153). Furthermore, there is a statistical distribution of particle spacings, and Arzt and Ashby suggest:

$$\tau_c/(Gb/L) = \alpha'/(1.68 + \alpha') \quad (154)$$

while Blum and Reppich use a similar relationship that includes the so-called Friedel correction:

$$\tau_c/(Gb/L) = \alpha'^{1.5} / (2\sqrt{2} + \sqrt{\alpha'^3}) \quad (155)$$

Equations (153)–(155), together with the values of α' , allow a determination of the threshold stress. Note that for general climb there is the suggestion that τ_c is particle-size independent.

The determination of α' will depend on whether climb is local or general; both cases are illustrated in Figure 83. The portion of the dislocation that climbs can be either confined to the particle matrix interfacial region (local), or the climbing region can extend beyond the interfacial region, well into the matrix (general). This significantly affects the α' calculation.

First, Eqn (153) suggests a maximum value for α' that corresponds to the Orowan bowing stress. It has been suggested that the Orowan stress can be altered based on randomness and elastic interaction considerations [556,559], giving values of $0.5 < \alpha' < 1.0$. For local climb, the value of α' depends on the shape of the particle, with $0.77 < \alpha' < 1.41$, from spherical to square shapes [555,556,559]. For extended or general climb, which is a more realistic configuration in the absence of any particular attraction to the particle [562], the α' is one order of magnitude or so smaller. Additionally, the value of α' will be dependent on the volume fraction, f , as $f^{1/2}$ and values of α' range from 0.047 to 0.14 from $0.01 < f < 0.10$ [538]. Blum and Reppich suggested that for these circumstances:

$$\text{for local climb, } \tau_c = 0.19[Gb/L]$$

$$\text{and for general climb, } \tau_c = 0.004 \text{ to } 0.02[Gb/L]$$

This implies that there is a threshold associated with the simple climbing of a dislocation over particle obstacles, without substantial interaction. This threshold stress is a relatively small fraction of the Orowan stress. There is the implicit suggestion in all of this analysis that the stress calculated from the above equations is athermal in nature, and this will be discussed subsequently.

2.3 Detachment Model

In connection with the above, however, there has been evidence that dislocations may interact with incoherent particles. This was observed by Nardone and Tien [563] and later by Arzt and Schroder [564] and others

[565] using transmission electron microscopy (TEM) of creep-deformed ODS alloys. Figures 85 and 86 illustrate this. The dislocations must undergo local climb over the precipitate and then the dislocation must undergo “detachment.” Srolovitz et al. [566] suggested that incoherent particles have interfaces that may slip and can attract dislocations by reducing the total elastic strain energy. Thus, there is a detachment stress that reflects the increase in strain energy of the dislocation on leaving the interface. Basically, Arzt and coworkers suggest that the incoherent dispersoids strengthen by acting as, essentially, voids. Arzt and coworkers [557,568–570] analyzed the detachment process in some detail and estimated τ_d as:

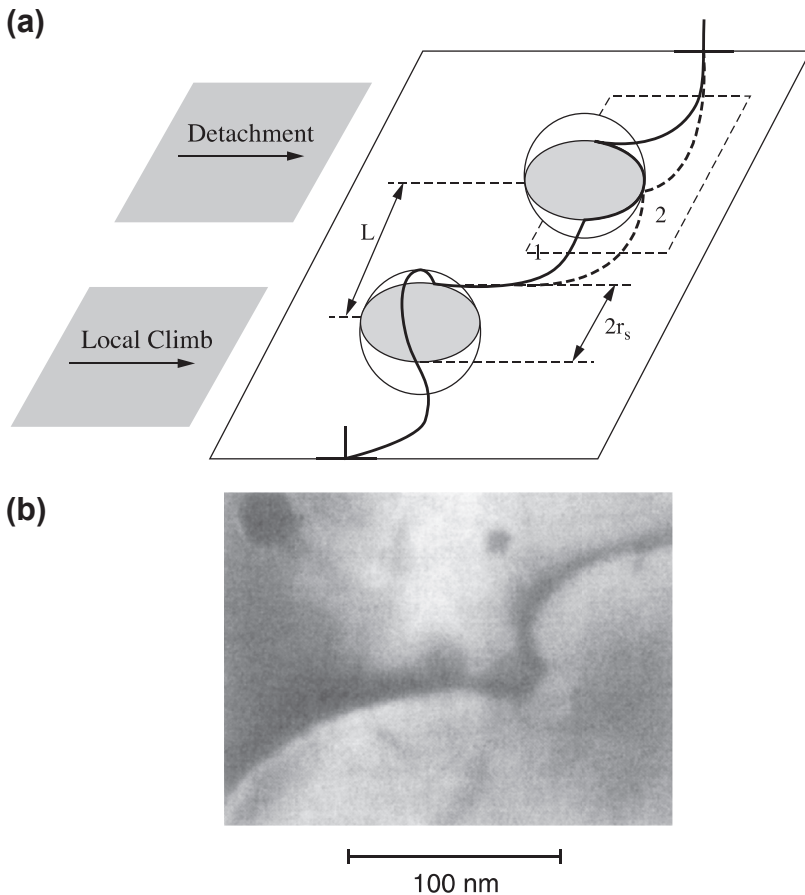


Figure 85 The mechanism of interfacial pinning. (a) Perspective view illustrating serial local climb over spherical particles of mean (planar) radius r_s and spacing L and subsequent detachment. (b) Circumstantial TEM evidence in the creep-exposed ferritic ODS superalloy PM 2000. From Ref. [565].

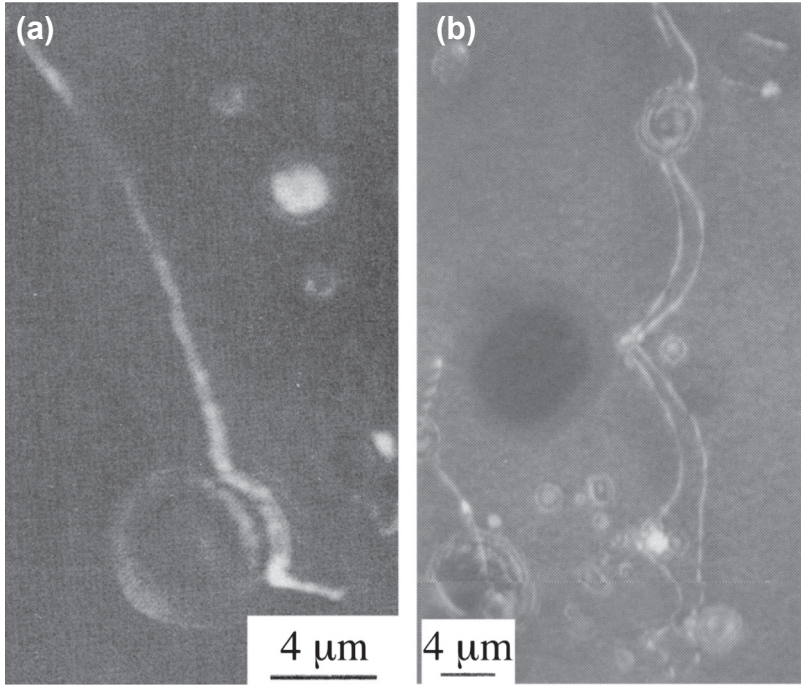


Figure 86 TEM evidence of an attractive interaction between dislocation and dispersoid particles. (a) Dislocation detachment from a dispersoid particle in a Ni alloy. (b) Dissociated superdislocation detaching from dispersoid particles in the intermetallic compound Ni_3Al . *From Refs [542,564,567].*

$$\tau_d = [1 - k_R^2]^{1/2} (Gb/L) \quad (156)$$

where k_R is the relaxation factor described by:

$$(Gb^2)_p = k_R (Gb^2)_m \quad (157)$$

where “ p ” refers to the particle interface and “ m ” the matrix. In the limit that $k_R = 1$, there is no detachment process.

Reppich [572] modified the Arzt et al. analysis slightly, using Fleischer–Friedel obstacle approximation, and suggested that:

$$\tau_d = 0.9(1 - k_R^2)^{3/4} / [1 + (1 - k_R^2)^{3/4}] (Gb/L) \quad (158)$$

This decreases the values of Eqn (156) by roughly a factor of 2. Note that as with general climb, τ_d and τ_c are independent of the particle size. It was suggested by both of the above groups that this detachment process

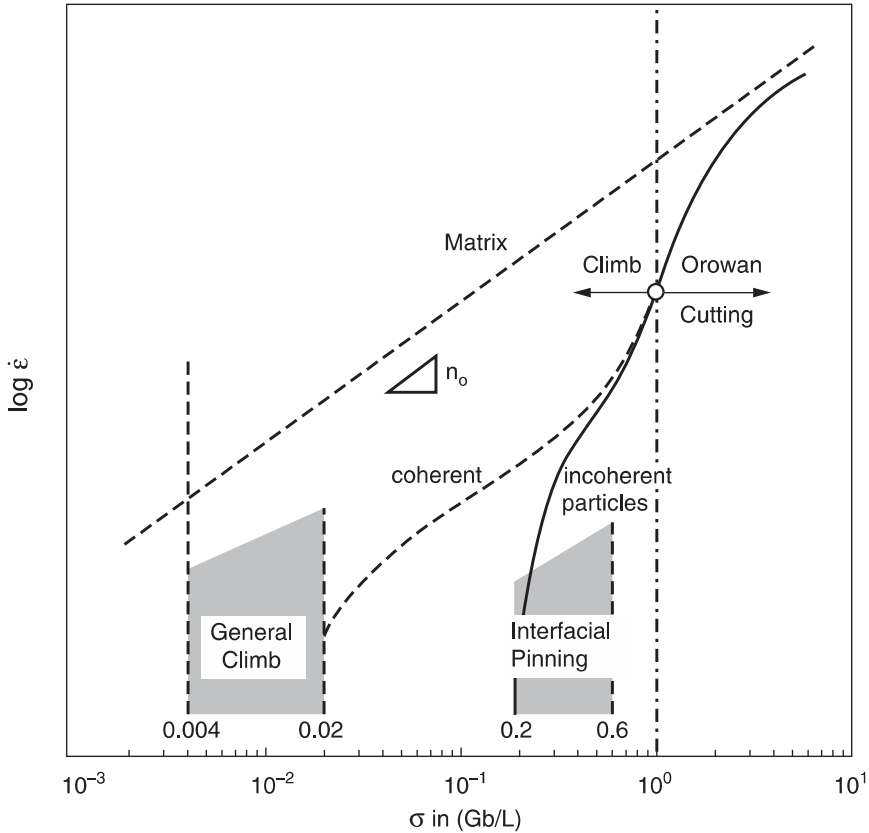


Figure 87 Creep behavior of particle-strengthened materials (schematic). The stress is given in units of the classical Orowan stress. From Ref. [538].

could be thermally activated. This equation suggests the τ_d is roughly $Gb/3L$, substantially higher than τ_c for general climb, as illustrated in Figure 87. Arzt and Wilkinson [557] showed that if k_R is such that there is just a 6%, or less, reduction in the elastic strain energy, then local climb become the basis of the threshold stress instead of detachment. For general climb, the transition point is k_R about equal to 1.

Rosler and Arzt [570] extended the detachment analysis to a “full kinetic model” and suggested a constitutive equation for “detachment-controlled” creep:

$$\dot{\epsilon} = \dot{\epsilon}_0 \exp \left[\left(-Gbr_s^2/kT \right) (1 - k_R)^{3/2} (1 - \sigma/\sigma_d)^{3/2} \right] \quad (159)$$

where $\dot{\epsilon}_0 = C_2 D_v L \rho_m / 2b$, and r_s is the particle radius. This was shown to be valid for random arrays of particles. Figure 88 plots this equation for

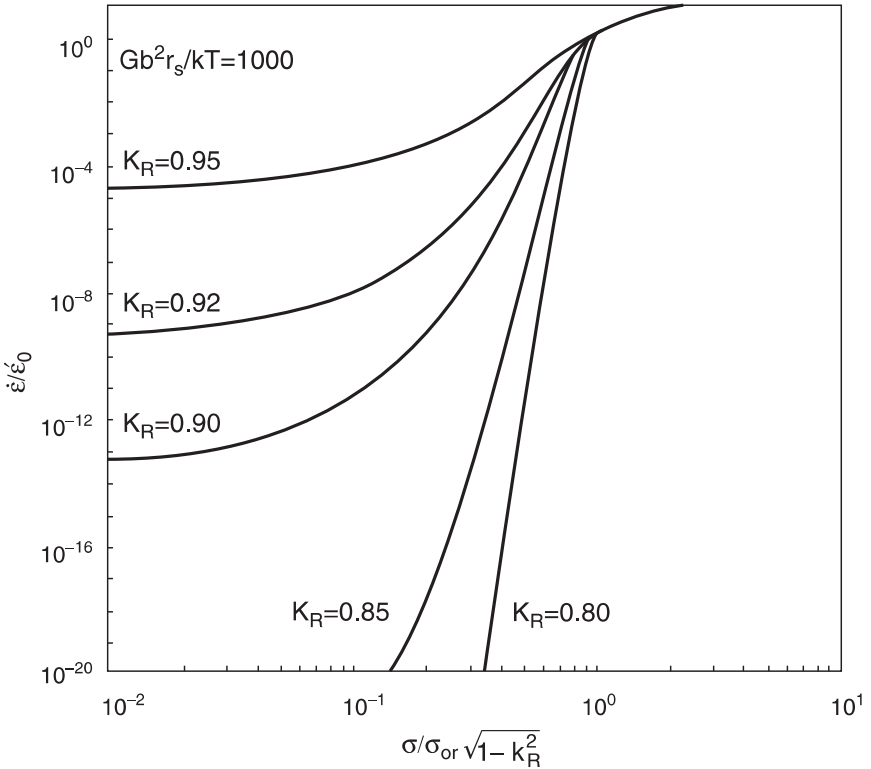


Figure 88 Theoretical prediction of the creep rate (normalized) as a function of stress (normalized) on the basis of thermally activated dislocation detachment from attractive dispersoids, as a function of interaction parameter k . The change of curvature at high strain rates (broken line) indicates the transition to the creep behavior of dispersoid-free material and does not follow from the equation. *From Ref. [569].*

several values of k_R as a function of strain rate. Threshold behavior is apparent for modest values of k_R . This model appears to reasonably predict the creep behavior of various dispersion-strengthened Al alloys [541,571] with reasonable k_R values (0.75–0.95). As Arzt points out, this model, however, does not include the effects of dislocation substructure. Arzt noted from this equation that an optimum particle size is predicted. This results from the probability of thermally activated detachment being raised for small dispersoids and that large particles (for a given volume fraction) have a low Orowan stress and, hence, small detachment stress [541]. It should be noted that Figure 88 does not suggest a “pure” threshold stress (below which plasticity does not occur). Rather, thermally activated detachment is suggested, and this will be discussed more later.

More recently, Reppich [540] reviewed the reported in situ straining experiments in ODS alloys at elevated temperatures and concluded that the observations of the detachment are essentially in agreement with the above description (thermal activation aside). In situ straining experiments by Behr et al., at 1000 °C in the TEM, also appear to confirm this detachment process in dispersion-strengthened intermetallics [567], as shown in Figure 86.

2.4 Constitutive Relationships

The suggestion of the above is that particle-strengthened alloys can be approximately described by relationships that include the threshold stress. A common relationship that is used to describe the steady-state behavior of second-phase strengthened alloys (at a fixed temperature) is:

$$\dot{\epsilon}_{ss} = A'(\sigma - \sigma_{th})^{n_m} \quad (160)$$

where σ_{th} is the threshold stress and n_m is the steady-state stress exponent of the matrix. Figure 89 graphically illustrates this superposition strategy. As will be discussed subsequently, this equation is widely used to assess the value of the threshold stress. Additional data that illustrates the value of Eqn (160) for superalloys is illustrated in Figure 90. Figure 91 illustrates that at higher stresses, above τ_{or} , decreases in stress (and strain-rate) illustrate a threshold behavior. That is, a plot of $\dot{\epsilon}^{1/n}$ versus σ extrapolates to τ_{or} , an “apparent σ_{th} .” However, as σ decreases below τ_{or} , a new threshold appears, and this then is the σ_{th} relevant to creep plasticity. A σ_{th} can be estimated with low stress plots such as Figure 91 [575,576] (also see Figure 87). These give rise to estimates of σ_{th} and allow plots such as Figure 90. However, Figure 89(b) and (c) illustrate that the high-temperature σ_{th} is not a true threshold and creep occurs below σ_{th} . This is why σ_{th} estimates based on plots such as Figure 91 decrease with increasing temperature.

An activation energy term can be included in the form of:

$$\dot{\epsilon}_{ss} = A'' \exp(-Q/kT) \left(\frac{\sigma - \sigma_{th}}{G} \right)^{n_m} \quad (161)$$

or

$$\dot{\epsilon}_{ss} = \frac{A'' D G b}{kT} \left(\frac{\sigma - \sigma_{th}}{G} \right)^{n_m} \quad (162)$$

where D is the diffusion coefficient. However, the use of D works in some cases while not in others for both coherent and incoherent particles.

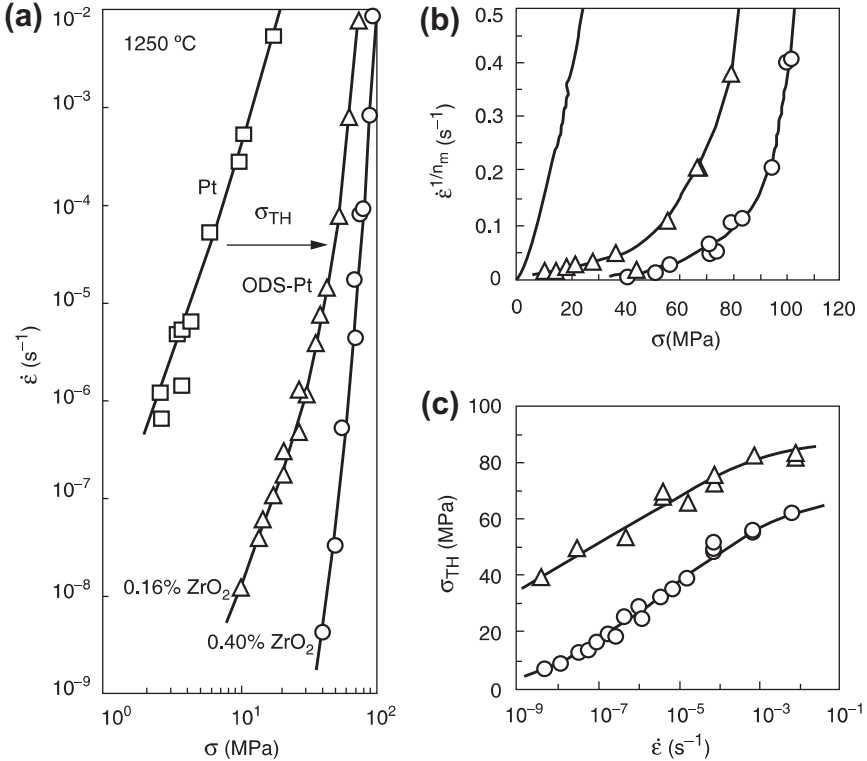


Figure 89 Comparison of the creep behavior of ZrO₂ dispersion strengthened Pt-based alloys at 1250 °C. (a) Double-logarithmic Norton plot of creep rate $\dot{\epsilon}$ versus stress σ . (b) Lagneborg-Bergman plot. (c) Dependence of σ_{th} on $\dot{\epsilon}$. From Ref. [567].

Figure 92 illustrates somewhat different behavior from Figure 82 in that the data do not appear to reduce to a single line when the steady-state stress is modulus-compensated and the steady-state creep rates are lattice-self-diffusion compensated. The activation energy for creep is reported to be relatively high at 537 kJ mol⁻¹ (as compared to 142 kJ mol⁻¹ for lattice self-diffusion). Cadek and coworkers [578,579] illustrated that for experiments on ODS Cu, the (modulus-compensated) threshold stress, determined by the extrapolation procedure described in Figure 92, is temperature-dependent. They propose that the activation energies should be determined using the usual equation but at constant $(\sigma - \sigma_{th})/G$ rather than σ/G as used in typical (especially five-power law for single-phase metals and alloys) creep activation energy calculations. The activation energies they calculated using this procedure reasonably correspond to

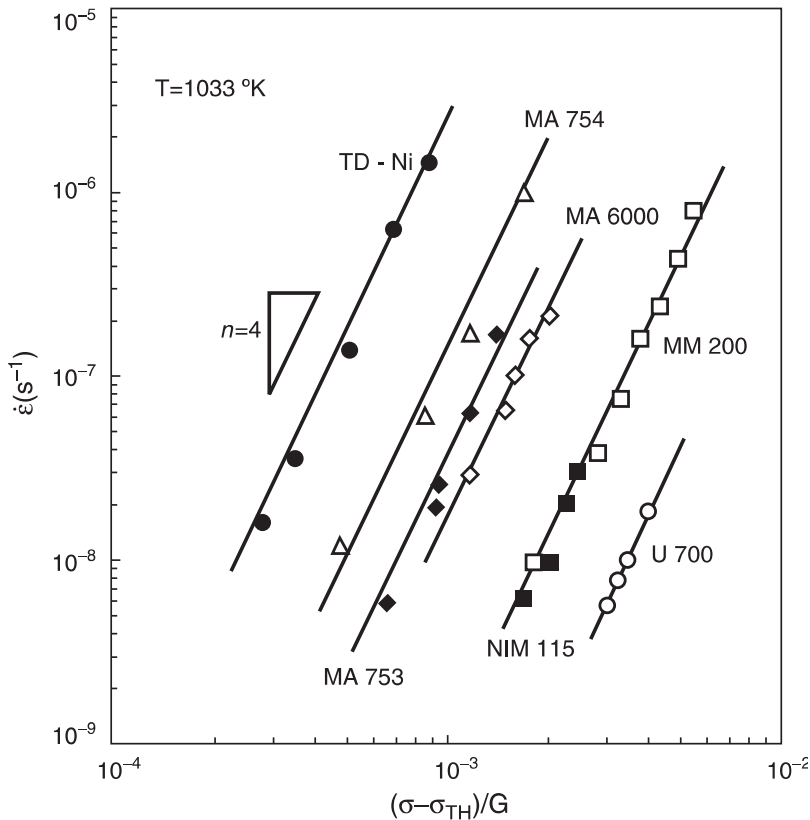


Figure 90 Double-logarithmic plot of creep rate versus reduced stress $\sigma - \sigma_{th}$ for various superalloys. After Ajaja et al. [573].

lattice or dislocation–core self–diffusion. Thus, the investigators argued that the activation energy for diffusion could reasonably be used as the activation energy term such as in Eqn (161).

Figure 87 is an idealized plot by Blum and Reppich that illustrates many of the features and parameters for particle strengthening. This is a classic logarithmic plot of the steady–state creep– (strain–) rate versus the steady–state stress. The value of σ_{or} is indicated and apparent threshold behavior is observed above this stress. A second threshold–like behavior is evident below the Orowan stress, one for incoherent particles and another for coherent particles at particularly low stresses (high temperature). The incoherent particles evince interfacial pinning and the more effective (or higher) threshold–like behavior is observed in the absence of a detachment

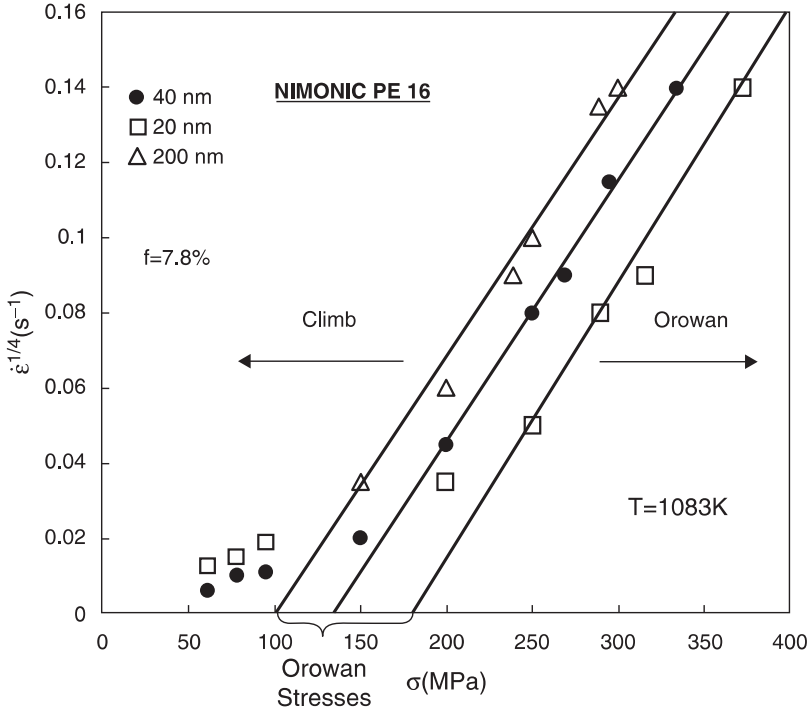


Figure 91 $\dot{\epsilon}^{1/n}$ versus σ -plot for determination of σ_{th} in γ' -hardened Nimonic PE 16 by back extrapolation (arrows). From Ref. [574].

stress. There has been some discussion as to what mechanism may be applicable at stresses below the apparent threshold, and it appears that grain boundary sliding and even diffusional creep have been suggested [580,581]. These, however, are speculative, as even single crystals appear to show sub-threshold plasticity. Figure 88 was basically an attempt to explain creep below the apparent σ_{th} through thermally activated detachment. This approach has become fairly popular [582], although recently it has been applied below, but not above, an apparent threshold. A difficulty with this approach is that Figures 85 and 86 suggest that a considerable length of dislocation is trapped in the interface, which would appear to imply a very large activation energy for detachment, much larger than that for D_v in Eqn (159). In at least some instances, the plasticity below the apparent threshold is due to a change in deformation mechanism [583]. Dunand and Jansen [584,585] suggested, for larger-volume fractions of second-phase particles (e.g., 25%), dislocation pile-ups become relevant and additional stress terms

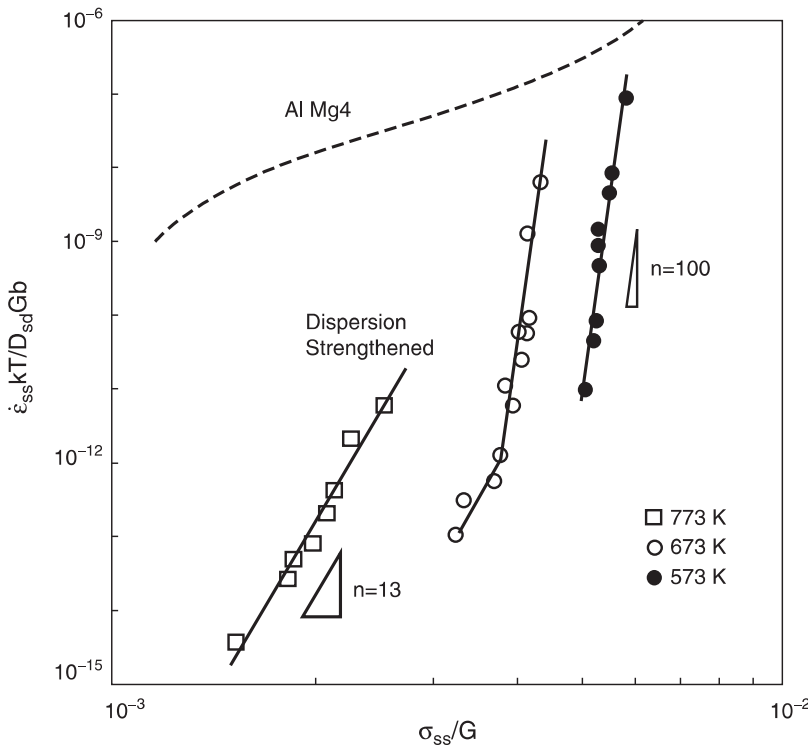


Figure 92 Lack of convergence of the different dispersion-strengthened creep curves with Q_{sd} compensation. From Ref. [577].

must be added to the conventional equations. However, these considerations do not appear relevant to the volume fractions being typically considered here.

2.5 Microstructural Effects

2.5.1 Transient Creep Behavior and Dislocation Structure

The strain versus time behavior of particle-strengthened alloys during primary and transient creep is similar to that of single-phase materials in terms of the strain rate versus strain trends as illustrated in Figure 93. Figure 93(a) illustrates Incoloy 800 H [544] and (b) Nimonic PE 16 [574]. Both generally evince Class M behavior although the carbide-strengthened Incoloy shows an inverted transient (such as a Class A alloy) but this was suggested to be due to particle structure changes, which must be considered with prolonged high-temperature application. The Nimonic alloy at the

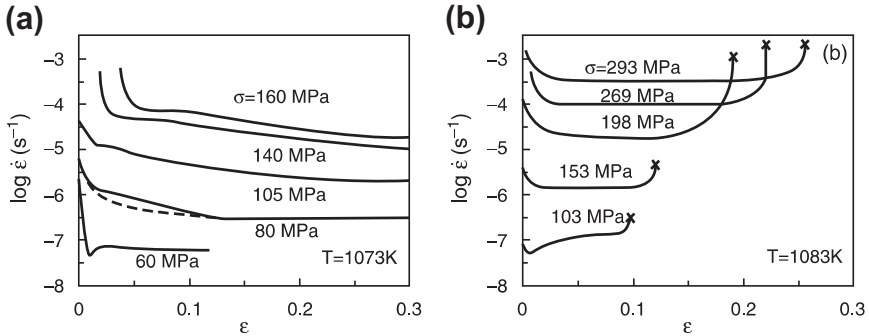


Figure 93 Half-logarithmic plot of creep rate $\dot{\epsilon}$ versus (true) strain of a single-phase material and particle-strengthened material. (a) Incoloy 800 H. (b) Nimonic PE 16. Adapted from Refs [538,574,586].

lowest stress also shows such an inverted transient and this was suggested as possibly being due to particle changes in this initially coherent γ' -strengthened alloy.

There has been relatively little discussed in the literature regarding creep transients. Blum and Reppich suggest that the transients between steady states with stress drops and stress jumps are analogous to the single-phase metals both in terms of the nature of the mechanical (e.g., strain rate vs strain) trends and the final steady-state strain-rate values, as well as the final substructural dimensions.

The dislocation structure of particle-strengthened alloys has been examined, particularly by Blum and coworkers [538,551]. The subgrain size in the particle-strengthened Al-Mn alloy are essentially identical to those of high-purity Al at the same modulus-compensated stresses. Similar findings were reported for Incoloy 800H [586] and TD-Nichrome [558]. These results show that the *total* stress level affects the subgrain substructure, even if there is an interaction between the particles and the subgrain boundaries. Blum and Reppich suggest, however, that the density of dislocations within the subgrains seems to depend on $\sigma - \sigma_{\text{th}}$ rather than σ , suggesting that particle hardening diminishes the network dislocation density compared to the single-phase alloy at the same value of σ_{ss}/G . Some of these trends are additionally evident from Figure 94. One interpretation of this observation is that the subgrain size reflects the stress but does not determine the strength. In contrast, Arzt suggests that only at higher stresses, where the alloys approach the behavior of dispersoid-free matrix, have dislocation substructures been reported [577]. Blum,

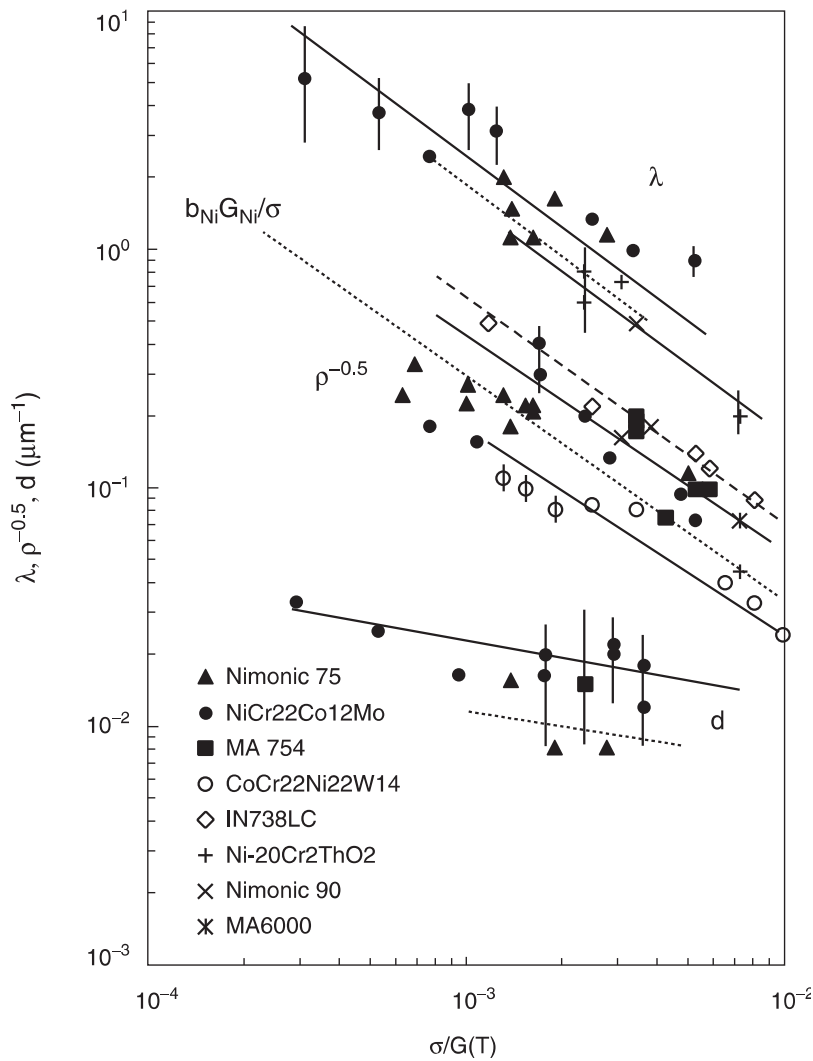


Figure 94 Steady-state dislocation spacings of Ni-base alloys. Adapted from Ref. [90].

however, suggests that this may be due to insufficient strain to develop the substructure that would ultimately form in the absence of interdiction by fracture [214].

2.5.2 Effect of Volume Fraction

As expected [580], higher-volume fractions, for identical particle sizes, are associated with greater strengthening and, of course, threshold behavior.

Others [587] have also suggested that the volume fraction of the second-phase particles can affect the value of the threshold stress.

2.5.3 Grain Size Effects

Lin and Sherby [548], Stephens and Nix [583], and Gregory et al. [511] examined the effects of grain size on the creep properties of dispersion strengthened Ni–Cr alloys and found that smaller grain size material may not exhibit a threshold behavior and evince stress exponents more typical of single-phase polycrystalline metals with high elongations [588]. Arzt [541] reports this sigmoidal behavior occurs in single crystals as well and the loss in strength (presumably below that of thermally activated detachment) involves other poorly understood processes including changes in the size or number of particles.

2.6 Coherent Particles

Strengthening from coherent particles can occur in a variety of ways that usually involve particle cutting. This cutting can be associated with (1) the creation of antiphase boundaries (e.g., γ – γ' superalloys), (2) the creation of a step in the particle, (3) differences in the stacking-fault energy between the particle and the matrix, (4) the presence of a stress field around the particle, and (5) other changes in the “lattice friction stress” [133].

Most of the earlier work referenced was relevant to incoherent particles. This is probably in part due to the fact that coherent particles are often precipitated from the matrix, as opposed to added by mechanical alloying, etc. Precipitates may be unstable at elevated temperatures and as a consequence, the discussion returns to that of incoherent particles. Of course, exceptions include the earlier referenced γ – γ' of superalloys and, more recently, the AlSc_3 precipitates in Al–Sc alloys by Seidman et al. [590,591]. In this latter work, coherent particles are precipitated. The elevated temperature strengths are much less than the Orowan bowing stress and also less than expected based on the shearing mechanism. Thus, it was presumed that the rate-controlling process is general climb over the particle, consistent with other literature suggestions. This, as discussed in the previous section, is associated with a relatively low threshold stress that is a small fraction of the Orowan bowing stress at about $0.03 \sigma_{\text{or}}$, and is independent of the particle size. Seidman et al. found that the normalized threshold stress increases significantly with the particle size and argued that this could only be rationalized by elastic interaction effects, such as coherency strain and modulus effects. Detachment is not important. The results are illustrated in

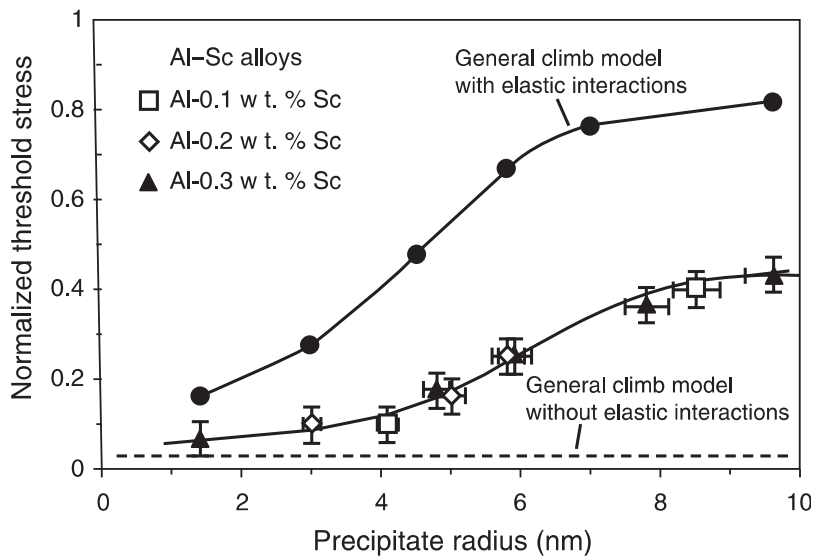


Figure 95 Normalized threshold stress versus coherent precipitate radius. From Ref. [591].

Figure 95. They also found that subgrains may or may not form. They do appear to obey the standard equations that relate the steady-state stress to subgrain size when they are observed. Seidman et al. do appear to suggest that steady state was achieved without the formation of subgrains.